

Topic 4: CSS Basics — Styling Web Pages

Learning Objectives

- Understand what CSS is and how to apply it to HTML
- Write CSS rules using selectors, properties, and values
- Style text, colors, backgrounds, and borders
- Understand CSS specificity and the cascade

4.1 What is CSS?

CSS (Cascading Style Sheets) controls the visual appearance of HTML elements — colors, fonts, spacing, layout, and more. Without CSS, every web page would look like a plain text document.

Three ways to add CSS:

```
<!-- 1. External stylesheet (recommended) -->
<link rel="stylesheet" href="styles.css">

<!-- 2. Internal style block (in <head>) -->
<style>
  h1 { color: blue; }
</style>

<!-- 3. Inline styles (avoid for large projects) -->
<h1 style="color: blue;">Hello</h1>
```

Always prefer **external stylesheets** — they separate HTML from CSS and allow one file to style an entire website.

4.2 CSS Rule Structure

```
selector {
  property: value;
  property: value;
}
```

Example:

```
h1 {
  color: #1d3557;
  font-size: 32px;
  font-weight: bold;
}

p {
  color: #333333;
  font-size: 16px;
  line-height: 1.6;
}
```

4.3 CSS Selectors

```
/* Element selector – targets all <p> tags */
p {
  color: gray;
}

/* Class selector – targets elements with class="highlight" */
.highlight {
  background-color: yellow;
}

/* ID selector – targets element with id="title" */
#title {
  font-size: 28px;
}

/* Multiple selectors at once */
h1, h2, h3 {
  font-family: Arial, sans-serif;
}

/* Descendant – only <a> inside <nav> */
nav a {
  text-decoration: none;
}

/* Hover state */
a:hover {
  color: blue;
}
```

```
text-decoration: underline;
}
```

4.4 Colors

```
/* Color name */
color: red;
color: darkblue;

/* Hexadecimal (most common) */
color: #ff0000;    /* red */
color: #1d3557;    /* dark navy blue */
color: #ffffff;    /* white */
color: #000000;    /* black */

/* RGB */
color: rgb(255, 0, 0);

/* RGBA (with transparency: 0 = invisible, 1 = fully visible) */
color: rgba(29, 53, 87, 0.5);
```

4.5 Text Styling

```
p {
  font-family: 'Arial', 'Helvetica', sans-serif;
  font-size: 16px;
  font-weight: bold;          /* normal, bold, or 100-900 */
  font-style: italic;
  text-decoration: underline;
  text-align: center;         /* left | right | center | justify */
  line-height: 1.6;           /* vertical spacing between lines */
  letter-spacing: 1px;        /* horizontal spacing between characters */
  text-transform: uppercase; /* lowercase | capitalize */
}
```

4.6 Backgrounds

```
div {
  background-color: #f0f0f0;
  background-image: url('banner.jpg');
  background-size: cover;      /* fill the entire container */
  background-position: center;
```

```
background-repeat: no-repeat;
}
```

4.7 Borders

```
div {
  border: 2px solid #333333;
  border-top: 4px solid #3a86ff; /* only top border */
  border-radius: 8px;           /* rounded corners */
  outline: 2px dashed red;      /* outline (outside border, no space) */
}
```

4.8 CSS Units

Unit	Description	Best For
`px`	Fixed pixels	Borders, images, fixed sizes
`%`	Relative to parent	Widths, responsive layouts
`em`	Relative to parent's font-size	Padding, margins
`rem`	Relative to root font-size	Font sizes (consistent)
`vw`	% of viewport width	Full-width sections
`vh`	% of viewport height	Full-height sections

4.9 The Cascade and Specificity

When multiple rules target the same element, **specificity** determines which one wins:

Priority order (highest to lowest):

1. Inline styles (`style="..."`) — always wins
2. ID selectors (`#title`)
3. Class selectors (`.highlight`)
4. Element selectors (`p` , `h1`)

```
/* All three rules target the same element */
/* <p class="intro" id="first">Hello</p> */

p      { color: gray; } /* least specific – loses */
```

```
.intro { color: blue; } /* more specific – overrides p */
#first { color: green; } /* most specific – wins */
```

Key Takeaways

- CSS controls visual appearance — colors, fonts, spacing, layout
- Use external stylesheets to separate CSS from HTML
- A CSS rule has a selector and a block of property-value pairs
- Hex color codes like `#1d3557` are the most common format
- Higher specificity (ID > class > element) wins when rules conflict

Reference Materials

- YouTube: Learn HTML & CSS Full Course (freeCodeCamp) — https://www.youtube.com/watch?v=a_iQb1lnAEQ
- MDN CSS Basics — https://developer.mozilla.org/en-US/docs/Learn/Getting_started_with_the_web/CSS_basics
- W3Schools CSS Tutorial — <https://www.w3schools.com/css/default.asp>

Quiz

Question 1: Which CSS property changes the text color of an element?

- A) `font-color`
- B) `text-color`
- C) `color`
- D) `foreground`

Correct Answer: C — The correct property is simply `color`. Properties like `font-color` and `text-color` do not exist in CSS.

Question 2: How do you write a CSS selector that targets an element with `id="title"`?

- A) `.title`
- B) `*title`
- C) `title`
- D) `#title`

Correct Answer: D — ID selectors use the `#` prefix (`#title`). Class selectors use `.` (`.title`). Element selectors use just the tag name (`p`, `h1`).
